

Subject: NT3 and functional nociceptor output

Hi Ela,

Thanks for sharing these follow-up results — they really add an important functional layer to the differentiation data.

From the optimisation experiment, 5 MOI NGN2 clearly maximised overall neuronal yield, and adding NT3 increased the proportion of MAP2+ cells. On that basis alone, 5 MOI + NT3 looked like the best condition for generating neurons.

Your results refine this picture. The TRKA immunostaining shows no significant difference in the proportion of TRKA+ cells within the MAP2+ population with or without NT3. So at the marker level, NT3 does not seem to meaningfully increase nociceptor specification.

However, the functional Ca²⁺ imaging data are more revealing. The proportion of mustard oil-responsive neurons is significantly higher in the absence of NT3. Given the strength of the p-value, this appears to be a robust effect. This suggests that although NT3 increases overall neuronal yield, it may reduce the fraction of neurons that are functionally nociceptive.

Putting both datasets together, we seem to have a trade-off:

- With NT3 → more total neurons (higher MAP2+ fraction), but a lower proportion of functionally responsive nociceptors.
- Without NT3 → fewer neurons overall, but a higher fraction of functional nociceptors.

Since the aim of the programme is to generate functional nociceptors for pain screening — not just any neurons — functional output is likely the more relevant endpoint. It's possible that the absolute number of functional nociceptors per well is comparable or even higher without NT3.

Based on the current data, I would keep 5 MOI NGN2 but reconsider including NT3 in the protocol. It would be useful to confirm absolute functional yields per well, but provisionally, omitting NT3 may give us a more suitable population for screening applications.

Best,
Parna